

Department of Physical Therapy
PTH6124: Clinical Inquiry I: Causality & Inference
Fall 2026

Course Information

Mode of Delivery: Hybrid (In-Person and Independent Study)
Credit Hours: 3
Semester Clock Hours: 37.5 (15 In-Person; 22.5 Independent Study)
Classroom: SRH 103

Class hours: Friday 8:00AM – 9:00AM
Weekly Clock Hours: 1

Instructor(s) and roles

Tim Duffy, PT, DPT
Office Hours: By Appointment
Tduffy1@plymouth.edu

Pre Requisites: Clin & Functional Anatomy, Clinical Phys, Ethics & Value Systems, Patient / Client Management I, Integrated Clinical I

Co Requisites: Neuroscience, Dynamic Systems I: Movement & Adaptation, Patient / Client Management II, Integrated Clinical II

Course Description

Clinical Inquiry I introduces students to knowledge-based practice and the role of research in developing scientific and theoretical foundations for and the critical analysis of physical therapy practice. The class covers the process of knowledge generation, and the use of causal models for knowledge representation. Students learn how to develop, evaluate and reason inferentially with graphical causal models (Bayesian networks), and how to apply these skills to learning, understanding and decision making.

Student Learning Outcomes (Course Objectives)

1. Explain the role of research in developing scientific and theoretical foundations of physical therapy practice and critical analysis of physical therapy practice. (7D9, 10, 11)
2. Explain the process of knowledge generation, and the use of causal models for knowledge representation. (7D9, 10, 11)
3. Define and explain dynamic inference as combination of causal, inductive, deductive, abductive, statistical and Bayesian inference in research and clinical practice. (7D9, 10, 11)
4. Explain causation and criteria applied when evaluating causal inference. (7D9, 10, 11)
5. Develop graphical causal models (directed acyclic graphs) that depict causal associations as a representation of knowledge for use in practice. (7D9, 10, 11)
6. Explain and apply graphical causal models to inferential reasoning associated with differential diagnosis, screening and intervention. (7D16)
7. Identify and evaluate the underlying causal model and its assumptions in a research study, including an evaluation of confounding and bias. (7D9, 10, 11)
8. Identify common biases used in arriving at decisions in situations of uncertainty. (7D9, 10, 11)
9. Explain the strengths and weaknesses of using simple heuristics during decision-making under uncertainty. (7D9, 10, 11)
10. Relate clinical inquiry and dynamic inference to the elements of the patient/client care management model including, but not limited to, how to evaluate data from the examination (7D16, 17, 18, 20, 21, 22)
11. Access and analyze the underlying causal assumptions in scientific literature. (7D9)
12. Identify and evaluate the best evidence for practice based on our understanding of and assumptions pertaining to the underlying causal structure with clinical judgment and patient/client values; needs; and preferences to determine the best care for a patient/client. (7D11)

Required Readings and Materials

- Portney LG. eds. Foundations of Clinical Research: Applications to Evidence-Based Practice, 4e. F. A. Davis Company; 2020.
- Other readings and handouts are provided on canvas as needed.

Free Software:

- DAGitty: <http://www.dagitty.net/>

Description of teaching methods and learning experiences

Lectures, Discussions, In class activities, online video presentations, problem sets, examinations

Methods of student evaluation/grading:

Assignments and Activities

Class Preparation and Participation (10 points)

Problem Sets (5 x 10 points each, 50 points total)

Final Examination (40 points)

Class Preparation and Participation

This course involves numerous in class activities such as data collection, building datasets and performing basic statistical calculations. Students earn credit based on being prepared for, and participating in these activities.

Exams

Final Examination – Includes all course content including content from all Problem Sets

Problem Sets (5)

Problem sets are completed to extend the in-class activities to individual work completing, interpreting, and writing up analyses using the data sets collected during in-class activities.

Problem Set 1 — Reasoning Under Uncertainty

Weeks 1–4: inference, causality, logical fallacies, cognitive biases, and clinical reasoning

Problem Set 2 — Statistical Inference: From Samples to Populations

Weeks 5–7: sampling variability, proportions, standard error, confidence intervals, and hypothesis testing

Problem Set 3 — Diagnostic Accuracy

Week 9: sensitivity, specificity, predictive values, accuracy, and interpretation of diagnostic tests

Problem Set 4 — Bayesian Clinical Reasoning

Weeks 10–12: probability, odds, likelihood ratios, and updating clinical beliefs

Problem Set 5 — Clinical Inquiry in Practice

Weeks 13–14: integrating causal, inductive, abductive, Bayesian, and deductive reasoning with human factors. This problem set serves as a **practice run for the final examination**, requiring students to integrate concepts from across the course to analyze clinical and research scenarios.

Student Evaluation / Grading Scale

A	94.0 – 100
A-	90.0 – 93.9
B+	87.0 – 89.9
B	83.0 – 86.9
B-	80.0 – 82.9
C+	77.0 – 79.9
C	73.0 – 76.9
F	below 73.0

Students are reminded that continuation in the program necessitates achieving an overall grade point average of 3.0 (B) or better. Grades below 2.0 (C) are not acceptable or counted toward the DPT degree. No more than 10 credits below 3.0 may be counted toward the DPT degree (grades of B-, C+, or C). Make-up examinations will be given only in specific cases of absence (documented absences). Circumstances that may warrant an absence should be discussed with the instructor before the scheduled time of the exam. Failure to notify the instructor as stated may result in a zero for the exam.

Professional Conduct: Students are reminded of the program expectation that they follow Professional Conduct, see the Student Handbook, Professional Conduct and Reasonable Professional Expectations for the process of handling violations, remediation and consequences.

Cell Phones: Cell phone use (including texting) is prohibited in class. Also, please respect your classmates by turning your phone to vibrate prior to class start time. Do not turn phones off as the University uses texts to relate emergency information.

Graduate Academic Policies: The DPT Program is a selective admissions cohort-structured program. Some Graduate Academic Policies may not be relevant to the program. Students should refer to the Doctor of Physical Therapy Program Handbook for specific information about Scheduling, Capstones, and Internships. The sections on Grading System and Academic Standing Policy do apply to DPT students: <https://coursecatalog.plymouth.edu/university-policies-procedures/graduate-academic-policies/>

Fair Grading Policy: Fair and equitable grading reflects values to which all members of the Plymouth State community commit themselves. Grades are used to assess the relative extent to which students achieve course objectives in all for-credit courses at PSU. Academic freedom allows instructors to determine course objectives within the bounds of established curricula, and the means by which a student's mastery of those objectives will be evaluated; and evaluate the quality of work on individual exams or assignments. Information can be found at <https://coursecatalog.plymouth.edu/university-policies-procedures/> under Fair Grading. <http://campus.plymouth.edu/faculty-governance/wp-content/uploads/sites/20/2017/05/PSU-Fair-Grading-Policy.pdf>

Student Support: Student Support Foundation (SSF) provides short-term emergency financial assistance and long-term student support. For more information, see: <https://campus.plymouth.edu/student-support-foundation/> Student Support Foundation also runs a food pantry, located in Belknap Hall. To learn more about SSF or access the food pantry, either via open hours or a private appointment, contact the SSF advisor, at psu-ssf@plymouth.edu.

Academic Dishonesty: Cheating and plagiarism are very serious forms of academic dishonesty. Any use of unauthorized assistance on exams, papers, homework assignments, or other course work constitutes cheating. Knowingly providing assistance during an exam or allowing other students to copy one's work is also a serious form of academic dishonesty. Plagiarism consists of submitting written work that has been developed wholly or partially by someone else. Submitting written work in which the ideas of others have been duplicated or even paraphrased without proper reference to the author is also a form of plagiarism. Also considered plagiarism is the acquisition of term papers or other assignments from another source and subsequent presentation of these materials as the student's own work. In addition, students may not use papers in more than one course without the permission of both instructors. Any student suspected of academic dishonesty will be reported to the Program Director immediately. A full description of academic integrity, violations and procedures when a violation to academic integrity in a course is suspected can be found at: <https://coursecatalog.plymouth.edu/university-policies-procedures/> under Academic Integrity.

Attendance Policy: Attendance is expected to all class sessions. Students are ultimately responsible for their attendance, however faculty will assess students' Professional Expectations based, in part, on their attendance (See Student Evaluation/Grading). Students who miss class for any reason are responsible for the information and work missed. Students should speak with the professor before missing the scheduled class when able.

Campus Accessibility Policy: Plymouth State University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you should immediately contact Campus Accessibility Services (CAS) in Speare 210 (535-3300) to determine whether you are eligible for accommodations. Academic accommodations will only be considered for students who have registered with CAS. If you have a Letter of Accommodation for this course from CAS, please provide the instructor with that information privately so that you and the instructor can review those accommodations:
<https://campus.plymouth.edu/accessibility-services/accommodations>.

LibGuide for Physical Therapy:

The Libguide for Physical Therapy is an information sharing platform created through a collaboration between the library and the DPT faculty that presents relevant resources to support students in the Physical Therapy program. Examples of information you will find include tutorials to better understand citing sources in APA and AMA style, resources on how to avoid plagiarism, quick access to relevant databases and interlibrary loan as well as a contact link to the program's assigned librarian. To view the DPT Program Libguide go to <https://library.plymouth.edu/physical-therapy>.

Diversity and Inclusivity Statement:

We consider this classroom to be a place where you will be treated with respect, and we welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, ability - and other visible and non-visible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class. See this web page for the Plymouth State University IDEA Center Inclusion, Diversity, Equity, and Access. <https://www.plymouth.edu/idea-center>

Tentative Class Schedule

Week	Date	Topic	Assignment Due
1	9/4	Why Do Smart Clinicians Disagree?	
2	9/11	Causality: Moving Beyond Association	
3	9/18	Causal Models: Directed Acyclic Graphs (DAGs)	
4	9/25	Human Decision Making: Heuristics, Biases, and Logical Fallacies	
5	10/2	From Samples to Populations:	Problem Set 1 Due
6	10/9	Sampling Distributions, Standard Error & Confidence Intervals	
7	10/16	Hypothesis Testing	Problem Set 2 Due
8	10/23	Self-Myofascial Release on Hamstring Flexibility	
9	10/30	Diagnostic Reasoning: Sensitivity, Specificity & Accuracy	Problem Set 3 Due
10	11/6	Bayesian Reasoning: Updating Beliefs	
11	11/13	Likelihood Ratios I: What Is a LR?	
12	11/20	Likelihood Ratios II: Using LRs in the Individual Patient	Problem Set 4 Due
—	11/27	THANKSGIVING — NO CLASS	
13	12/4	Human Factors I: Humans in Research	
14	12/11	Human Factors II: Humans in Clinical Practice	
15	12/18	Final exam review	Problem Set 5 Due